

Pseudocode Is a way to write solution using English language statements	Algorithm Step by step procedure to solve a problem using variables, constant and operators	Code (programming language)
1.Input first number 2.Input second number 3.Add first and second number together 4.Output the answer	1. input a 2. input b 3. c=a+b 4. print c	<pre> a=int(input("ENTER first NUMBER---->")) b=int(input("ENTER second NUMBER---->")) c=a+b print ("Result is",c) </pre>

Data types

Int----- 7,-5

Real/Float---- 8.4, 6/4

Text----- Pakistan

String----- Muhammad.waqar@77gmail.com

Boolean ---- true /false

Char ---- 'm'

CONDITIONAL STATEMENTS

There are two types of conditional statement as shown below with an example of each.

1. IF ... THEN ... ELSE ... ENDIF

- A condition that can be true or false;, for example

```
IF Age < 18
THEN PRINT "Child"
ELSE PRINT "Adult"
ENDIF
```

2. CASE ... OF ... OTHERWISE ...ENDCASE

- A choice between several different values;, for example

```
CASE Grade OF
"A" : PRINT "Excellent"
"B" : PRINT "Good"
"C" : PRINT "Average"
OTHERWISE PRINT "Improvement is needed"
ENDCASE
```

IF ... THEN ... ELSE ... ENDIF

For an IF condition the THEN path is followed if the condition is true and the ELSE path is followed if the condition is false. There may or may not be an ELSE path. The end of the statement is shown by ENDIF.

Structure

IF (condition)

then

```
{
  Value if true
}
```

else

```
{
  Value if false
}
```

endif

Algorithm # 1

Write an algorithm that input two numbers and output the sum.

Input a,b

sum=a+b

Output sum

Algorithm # 2

Write an algorithm that input three numbers and output the sum and average of these numbers.

Input a,b,c

sum=a+b+c

avg=sum/3

Output sum

Algorithm # 3

Write an algorithm that input amount in dollars and convert it into rupees.

Input dollar

rupees=dollar*160

Output rupees

Algorithm # 4

Write an algorithm that input a number and calculate its square and cube.

Input a

square = $a * a$

cube = square * a

Output square, cube

Algorithm # 5

Write an algorithm that input a number and output whether it is positive or negative.

Pseudocode

Input a

If $a > 0$

then

 print "Positive"

else

 print "Negative"

Python code

```
a=int(input("Enter a number--->"))
```

```
if a>0:
```

```
    print("Positive")
```

```
else:
```

```
    print ("Negative")
```

Algorithm # 6

Write an algorithm that input two numbers and output the greater number.

Pseudocode

Input a,b

If $a > b$

then

 print a

else

 print b

Python code

```
a=int(input("Enter first number--->"))
```

```
b=int(input("Enter second number--->"))
```

```
if a>b:
```

```
    print("A is greater",a)
```

```
else:
```

```
    print ("B is greater",b)
```

Algorithm # 7(Nested if)

Write an algorithm that input three numbers and output the greatest number.

Pseudocode/Algorithm

Input a,b,c

If (a>b AND a>c)

then

 print a

elseif (b>a AND b>c)

 print b

else

 print c

Python code

```
a=int(input("Enter first number--->"))
```

```
b=int(input("Enter second number--->"))
```

```
c=int(input("Enter third number--->"))
```

```
if a>b and a>c:
```

```
    print("A is greatest",a)
```

```
elif b>a and b>c:
```

```
    print ("B is greatest",b)
```

```
else:
```

```
    print("C is greatest",c)
```

Python code

```
a=int(input("Enter first number--->"))
```

```
b=int(input("Enter second number--->"))
```

```
if a>b:
```

```
print("A is greater",a)
else:
    print ("B is greater",b)
```

Algorithm # 8

Write an algorithm that input a number and output whether it is even or odd.

Pseudocode

Input a

If $a \% 2 == 0$ [$==$ relational operator , $\%$ - modulus sign returns remainder]

Print "Number is Even"

Else

Print "Number is Odd" $5 \% 2 = 1$ || $5 / 2 = 2.5$

Python Code

```
a=int(input("Enter a number--->"))
```

```
if a%2 == 0:
```

```
    print("Number is EVEN")
```

```
else:
```

```
    print ("Number is Odd")
```

Nested If (multiple else if)

Algorithm # 9

Write an algorithm that input percentage of student and output the grade.

A*= 90+ A=80-90 B=70-79 C=60-69 D=50-59 E=40-49 U=<40

Pseudocode

Input per

If per > 90 **then** print "Grade is A*"

else if per>=80 AND per<=90 then print "Grade is A"

else if per>=70 AND per<=79 then print "Grade is B"

else if per>=60 AND per<=69 then print "Grade is C"

else if per>=50 AND per<=59 then print "Grade is D"

else if per>=40 AND per<=49 then print "Grade is E"

else

print "Grade is U"

Python code

```
per=int(input("ENTER PERCENTAGE:"))
```



```
if per>90:
    print("GRADE IS A*")
elif per>=80 and per<=90:
    print("GRADE IS A")
elif per>=70 and per<80:
    print("GRADE IS B")
elif per>=60 and per<70:
    print("GRADE IS C")
elif per>=50 and per<60:
    print("GRADE IS D")
elif per>=40 and per<50:
    print("GRADE IS E")
else:
    print("GRADE IS U")
```

Loops

loop is a technique that repeat itself for specified number of times

1. Counter controlled loop(FOR-NEXT)
2. Condition controlled loop(1.WHILE..DO..ENDWHILE ,
2.REPEAT..UNTIL..ENDREPEAT)

1.Counter controlled loop

Input 26 numbers

Input a,b,c,d,e,f,g,h,i,j,k,.....,z X

FOR...NEXT LOOP

For x= 1 to 10

[

]

Next x

Algorithm # 10

Write an algorithm that input ten numbers and output the highest.

Algorithm 6, 3, 7, 2, 8, 9, 5, 2, 1, 7

h=0

for x = 1 to 10

input m

if m > h

h ← m

print h

next x

Python code

h=0

for x in range(1,11):

m=int(input("ENTER MARKS:"))

if m>h:

```
h=m  
print ("Highest marks is ",h)
```

Algorithm # 11

Write an algorithm that input a number and output the table.

Algorithm

Input n

For i = 1 to 10

r = n * i

print (n,"X",i,"=",r)

next i

python Code

```
n=int(input("ENTER NUMBER:"))
```

```
for i in range(1,11):
```

```
    r=n*i
```

```
    print (n,"X",i,"=",r)
```

Algorithm # 12

Write an algorithm that input 20 numbers and output whether how many are positive and how many are negative.

Pseudocode

```
pos=0 ✓  
neg=0 ✓  
For x= 1 to 20 ✓  
  Input num -4 7  
  If num>0  
  then  
    1 pos=pos+1 0+1 (1+1)  
  else  
    1 neg=neg+1 0+1  
next x  
print pos,neg
```

Python code

```
pos=0  
neg=0  
for x in range(1,21):  
    num=int(input("ENTER A NUMBER:"))  
    if num>0:  
        pos=pos+1  
    else:  
        neg=neg+1  
print("Positive=",pos,"Negative=",neg)
```

Algorithm # 13

Write an algorithm that input 20 numbers and output whether how many numbers are even and how many are odd.

Pseudocode

```
even=0
odd=0
For x= 1 to 20
Input num
If num % 2 == 0
then
    even=even+1
else
    odd=odd+1
next x
```

Print even,odd

Python Code

```
even=0
odd=0
for x in range (1,21):
    num=int(input("ENTER A NUMBER:"))
    if num%2==0:
        even=even+1
    else:
        odd=odd+1
print ("Even=",even,"Odd=",odd)
```